

Cellular Automata in Materials Science Applications

Hosted by the Wang Materials Group



The University of Texas at Austin
McKetta Department
of Chemical Engineering
Cockrell School of Engineering



Today's plan

What is Materials Science and Engineering?

Materials Science Tetrahedron

What is computation and simulation?

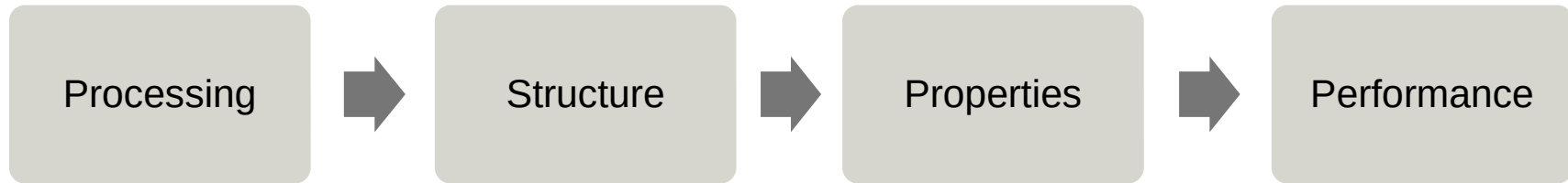
What is cellular automata?

Conway's Game of Life

Materials Science Example: Spinodal decomposition

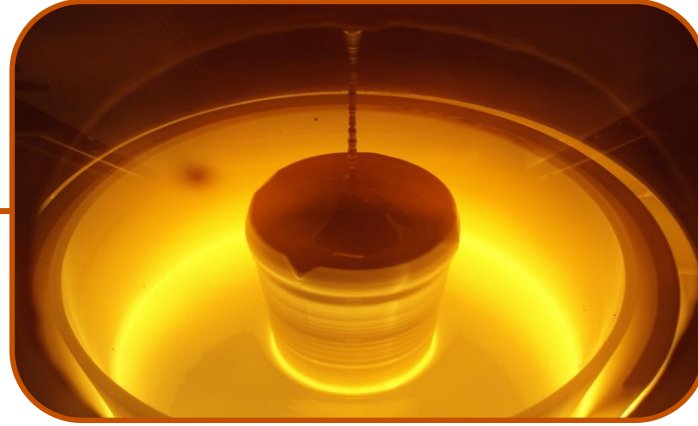
What is materials science and engineering?

Discovering, understanding, and designing materials (often times solids) through the study and application of processing-structure-property-performance relationships



What is materials science and engineering?

How is the material actually made in a lab/factory or adapted from nature?



Processing



Structure

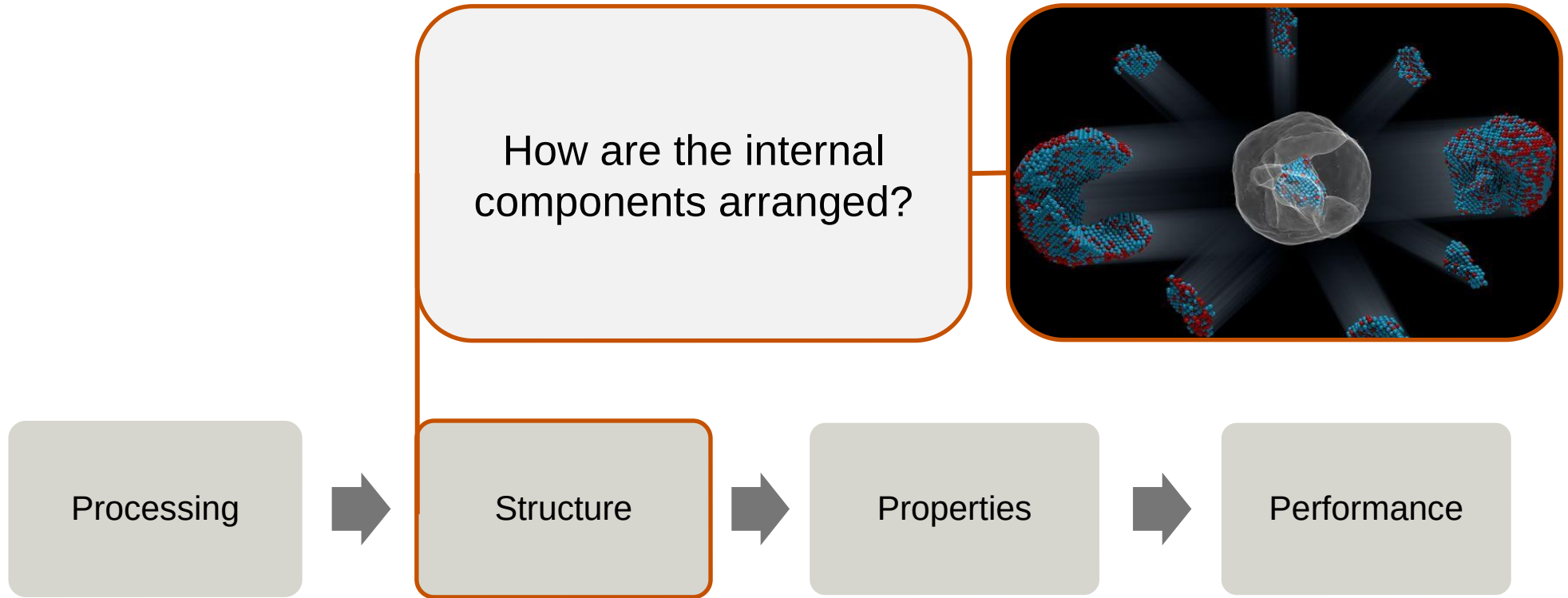


Properties

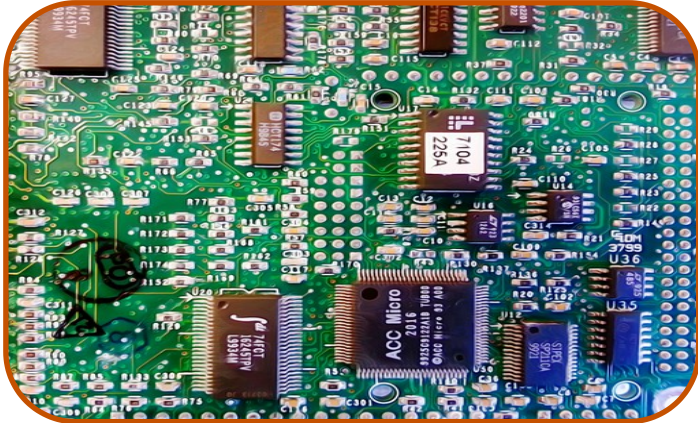


Performance

What is materials science and engineering?



What is materials science and engineering?



How does it respond to an external stimulus?

Mechanical

Electrical

Thermal

Magnetic

Optical

Chemical

Processing



Structure

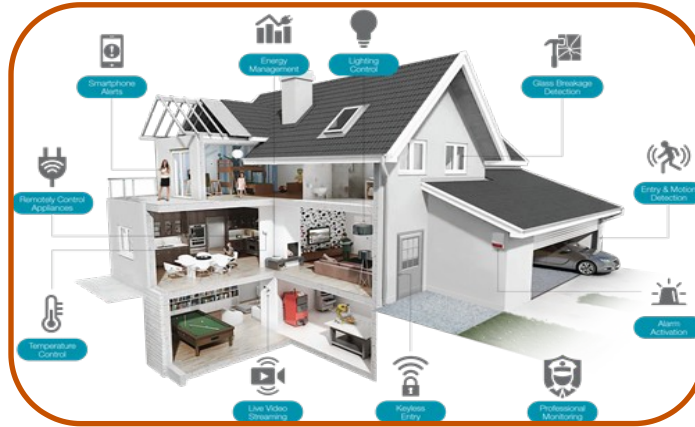


Properties



Performance

What is materials science and engineering?



How does the material help the user/world?

Processing



Structure



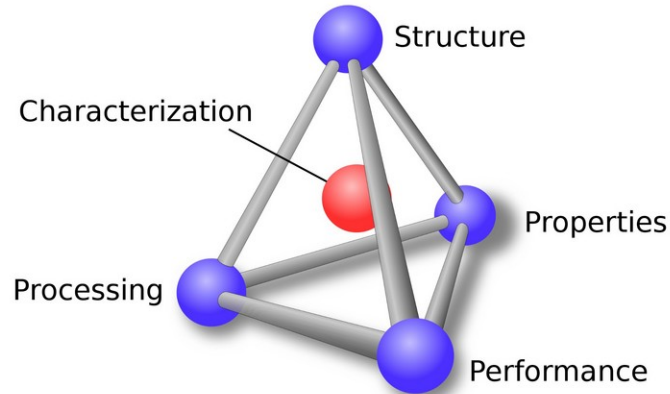
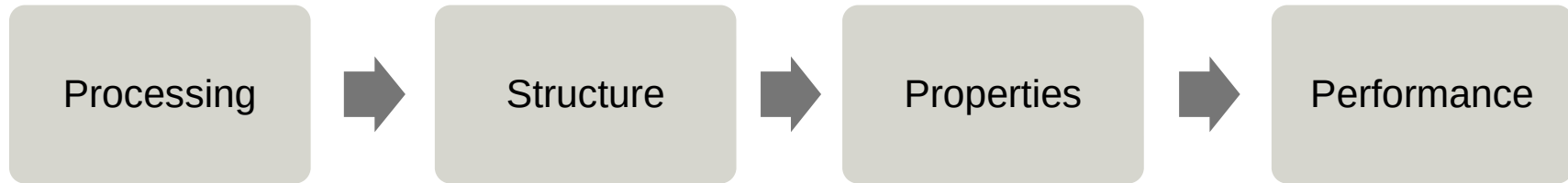
Properties



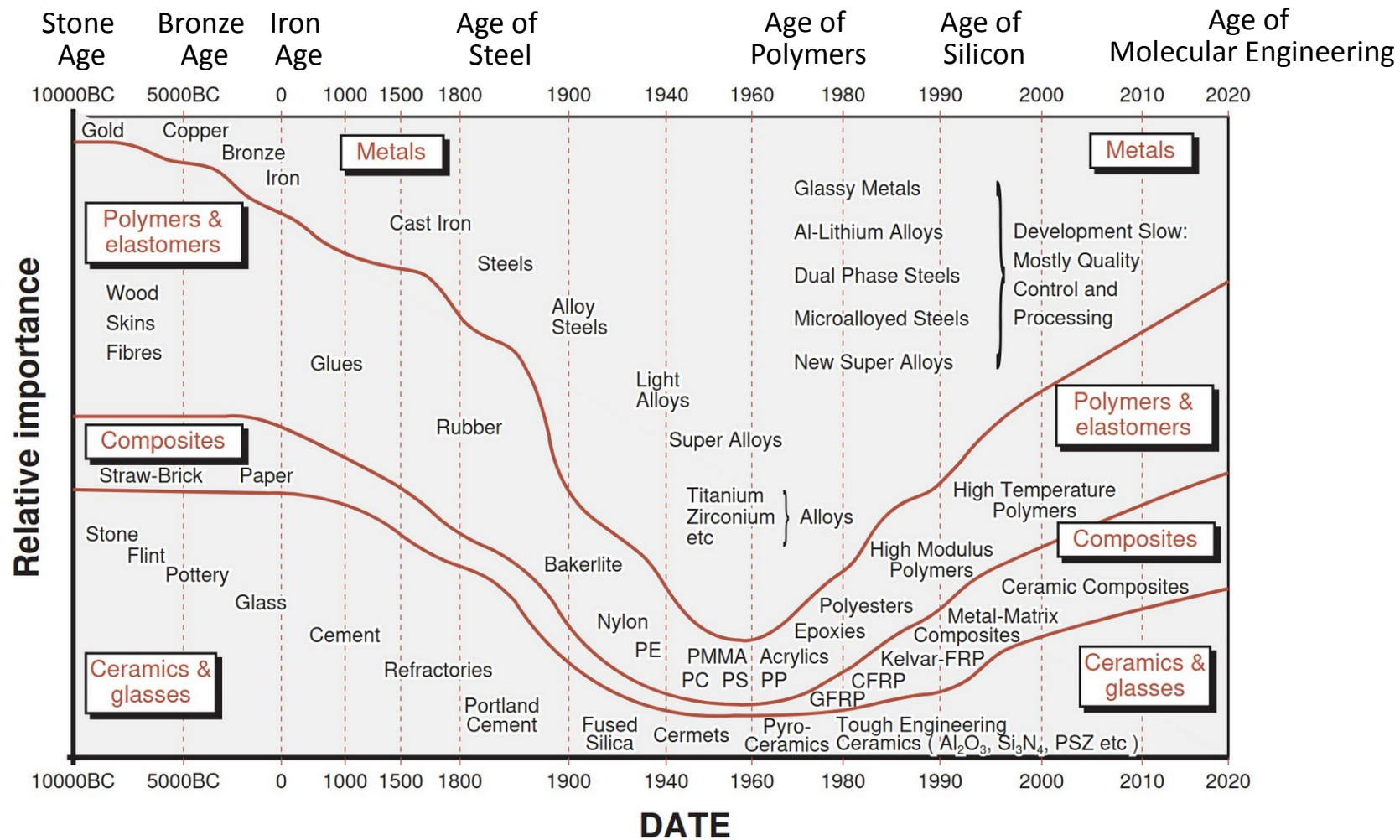
Performance

What is materials science and engineering?

Discovering, understanding, and designing materials (often times solids) through the study and application of processing-structure-property-performance relationships



Why study materials? (Think-pair-share)



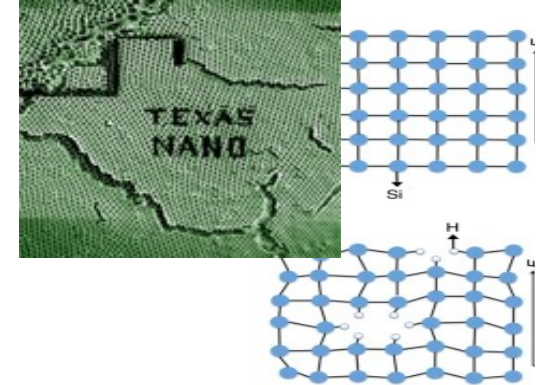
Examples of Materials Science and/or Engineering



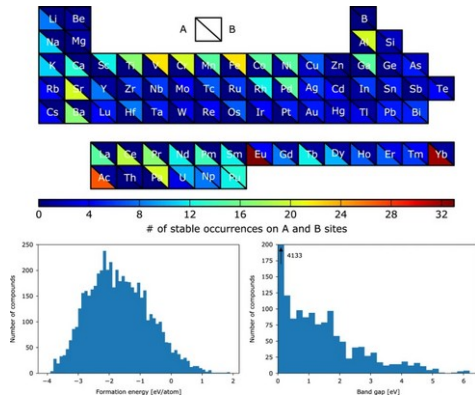
3D printing soft bio-polymers
Nelson Lab, University of Washington



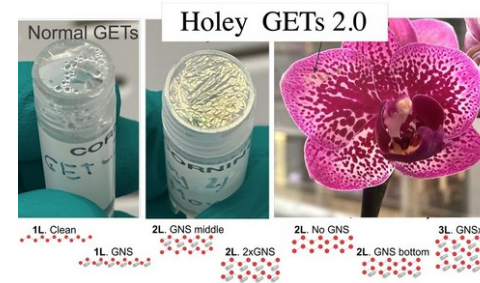
Using viruses as templates to
make more efficient solar cells
Belcher Lab, MIT



Using defects in nanoparticles for high-efficiency, low-
loss nanophotonics
Korgel & Zheng Research Groups, UT Austin

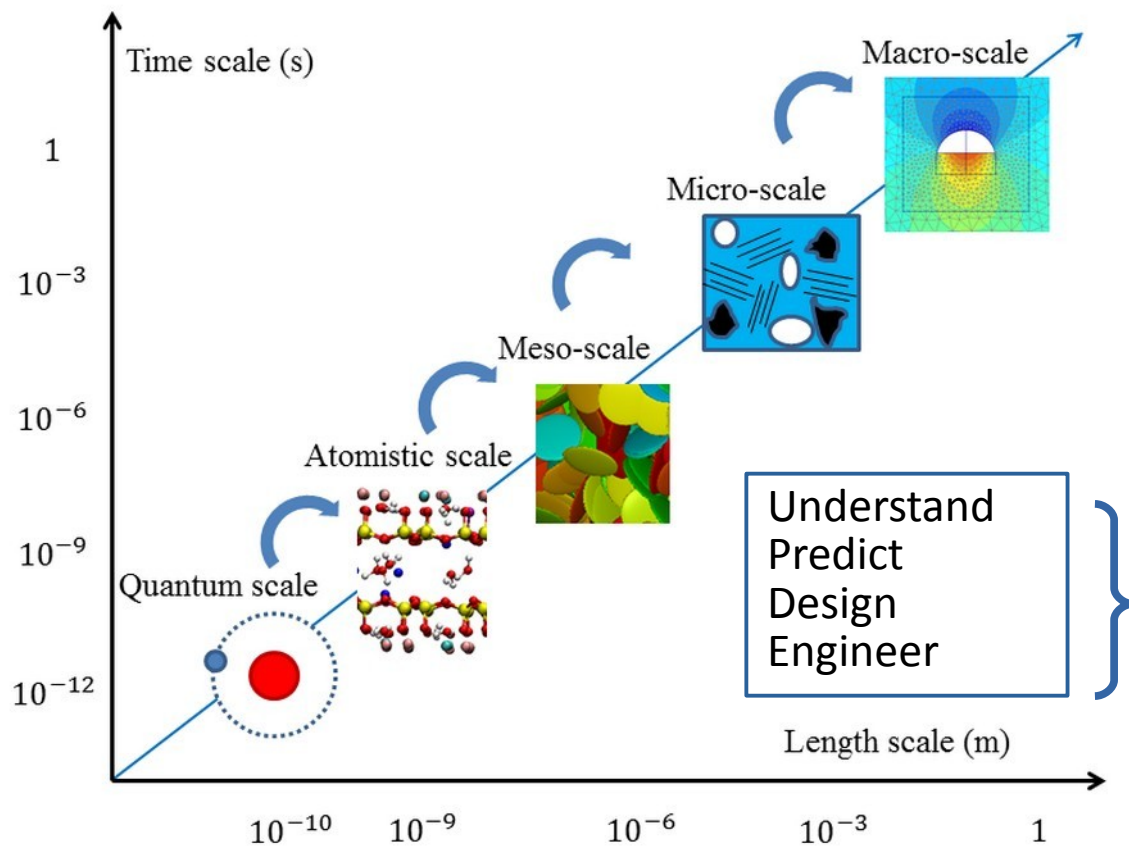


Using supercomputers to discover
new and ideal oxides
Wolverton Research Group, Northwestern



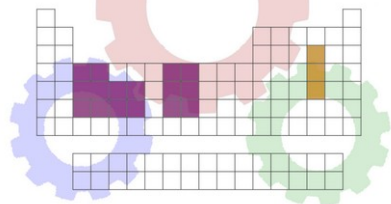
Breathable graphene tatoos
for personalized healthcare
Akinwande Nano Research Group, UT Austin

What is Modeling and Simulation in Materials Science?



Research Overview and Philosophy

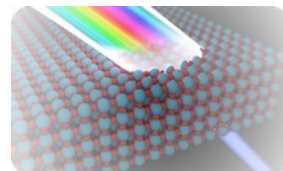
**AI-informed materials
discovery**



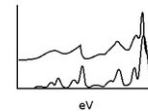
First-principles calculations



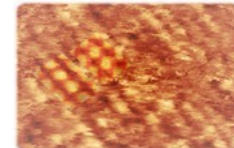
HPC resources



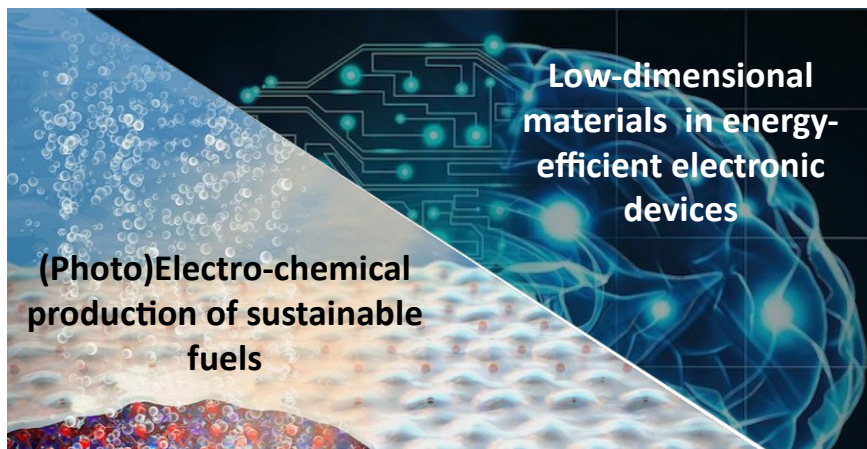
Experimental collab.



Spectroscopies, microscopies, ...



**Harnessing Materials Imperfections
for Energy Sustainability**



Conway's Game of Life

Variation of cellular automata

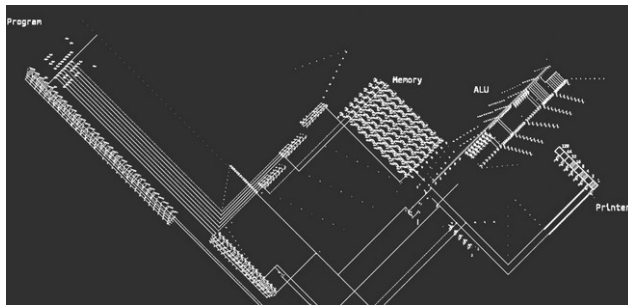
Invented by Cambridge mathematician John Conway (~1969)

“He spent most of his time playing games, inventing games and reinventing rules to games he found boring, such as chess.”

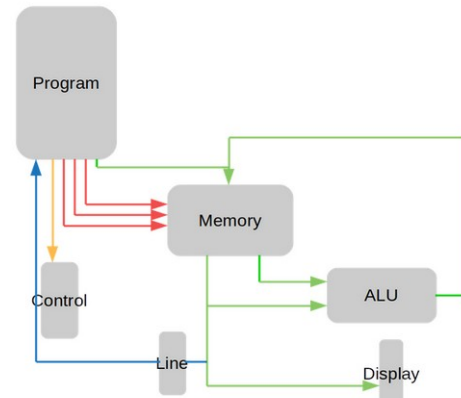
[The Guardian \(July 15, 2023\)](#)



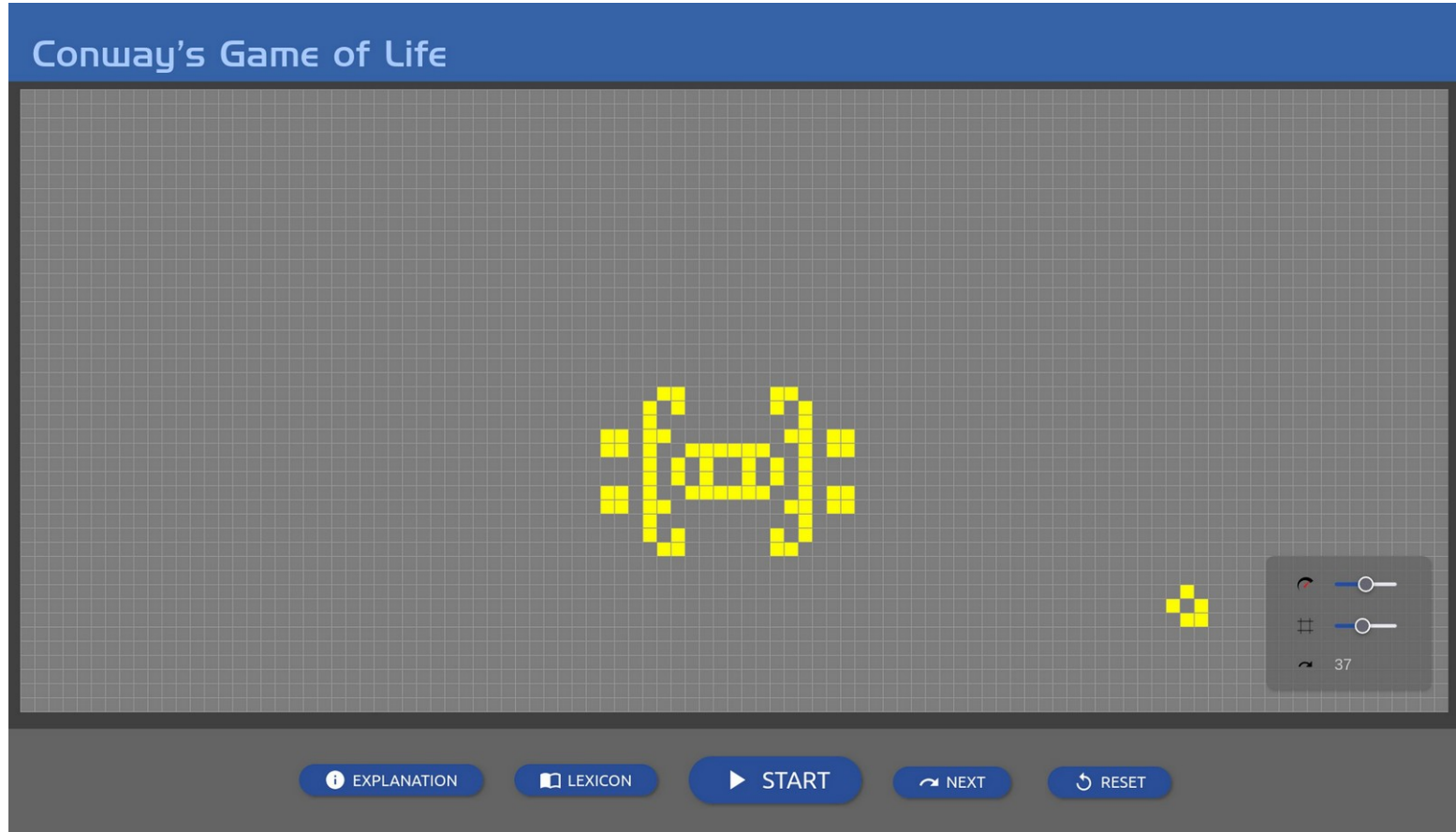
“no-player never-ending” game
Known to be Turing-complete



<https://www.nicolasloizeau.com/gol-computer>



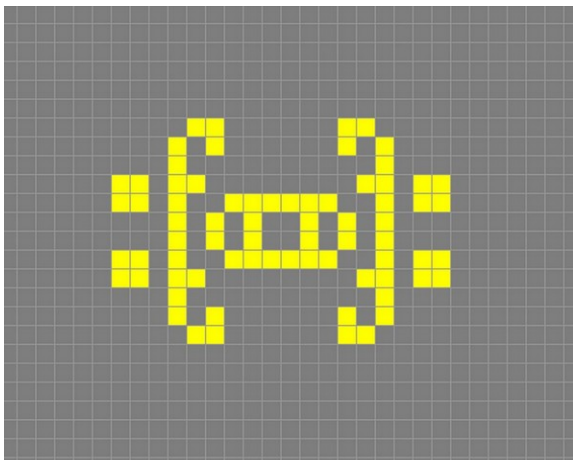
Conway's Game of Life



<https://playgameoflife.com/>

Conway's Game of Life

On a 2D square grid, at $t=0$, some cells are 'alive' (yellow) and the rest are 'dead' (grey).



At each time step, the value of the cell is updated:

- Alive cells stay alive if they have **two** or **three** living neighbors.
- Dead cells are revived to alive cells if there are **exactly three** living neighbors (imitating reproduction).
- Alive cells die if there are **fewer than two** living neighbors (underpopulation) or **more than three** living neighbors (overpopulation).

A more realistic example: Spinodal Decomposition in Alloys

Alloys



Brass (Cu, Zn, ...)



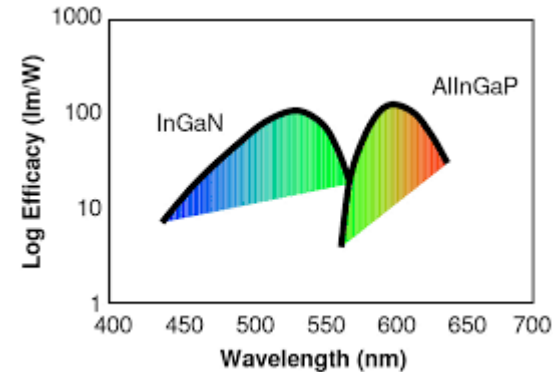
Bronze (Cu, Sn,...)



Steel (Fe, C,...)

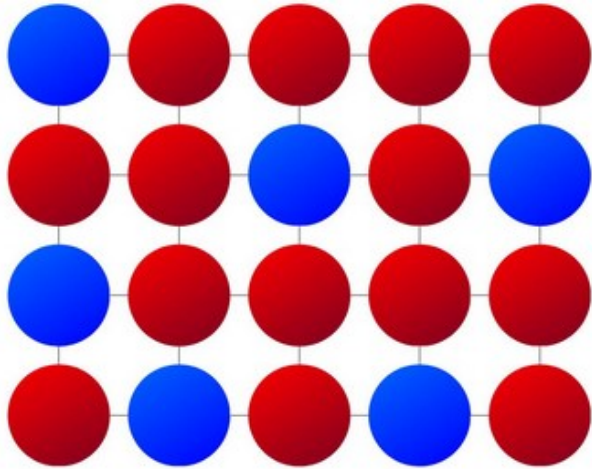


InGaN LEDs

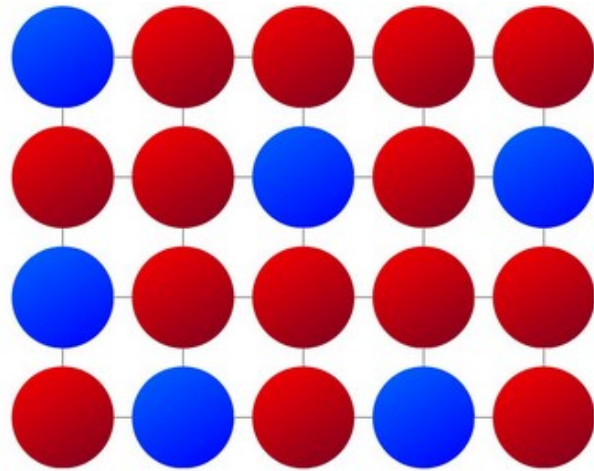


A more realistic example: Spinodal Decomposition

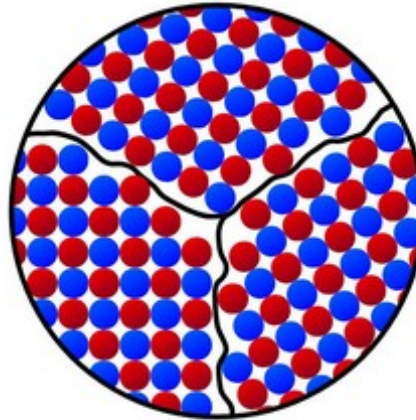
Alloys



Many alloys are not perfect solid solutions



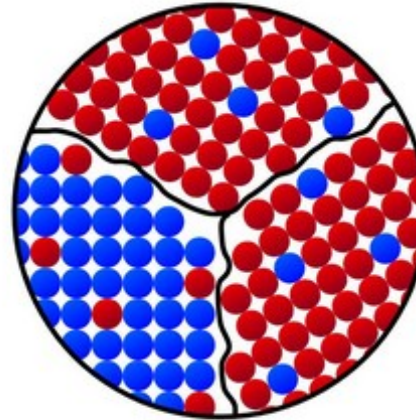
completely soluble



solid solution

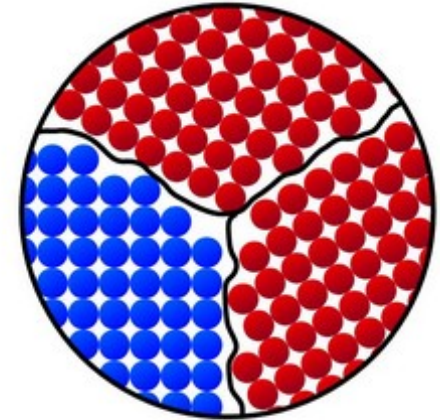
types of alloys
|
the components of the alloy are
in the solid state ...

partially soluble



mixture of
solid solutions

completely insoluble



mixture of
pure crystallites

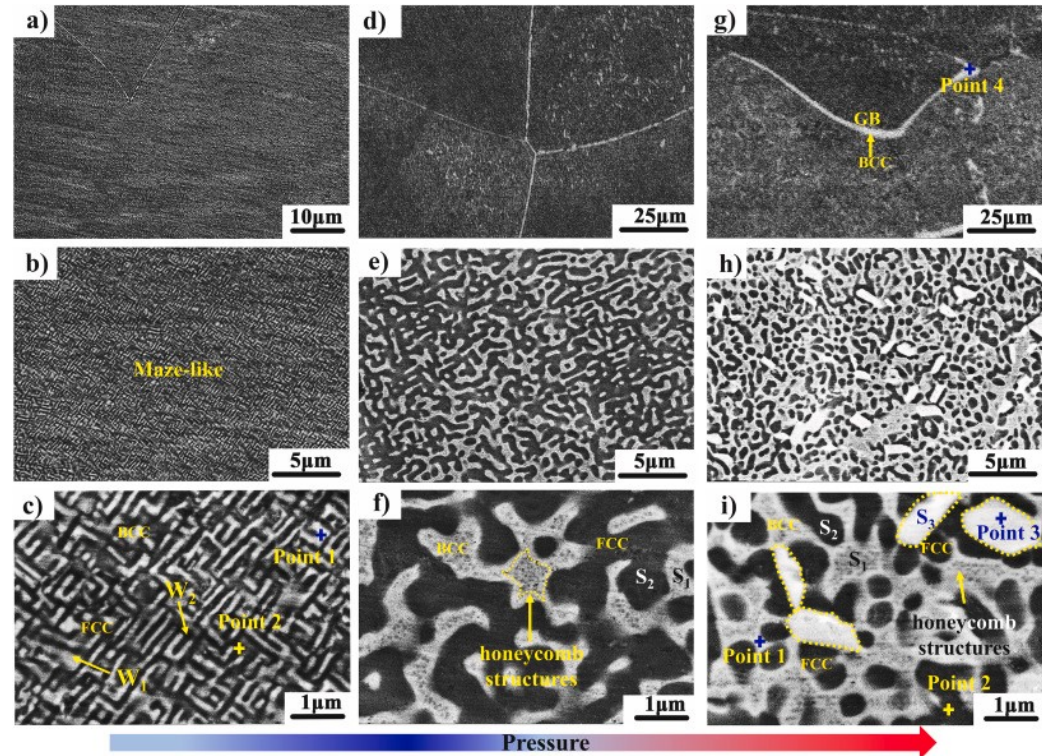
Mixture of “phases”

Spinodal Decomposition

Spinodal Decomposition

is spontaneous phase separation below a critical temperature,
often limited by diffusion

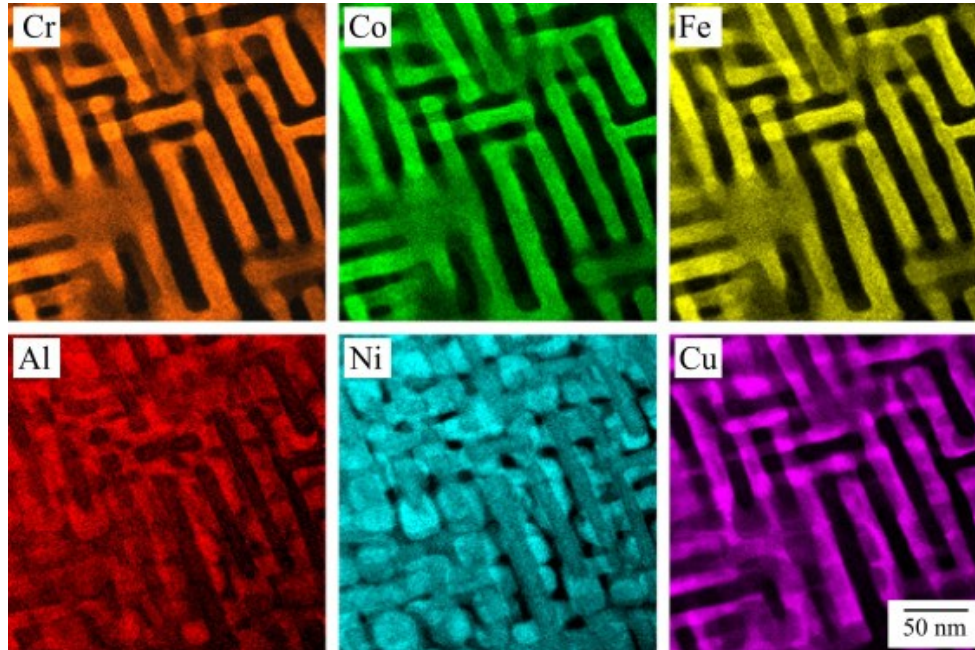
Microstructure evolution with
pressure for CoCrFeNiAl alloy



Spinodal Decomposition

Spinodal Decomposition

is spontaneous phase separation below a critical temperature,
often limited by diffusion



STEM-EDX image of spinodal decomposition of high-entropy alloy $\text{Al}_{0.5}\text{CrFeCoNiCu}$

A brief intro to Python

Python is a high-level, general-purpose programming language.

Its design philosophy emphasizes code readability.

This makes Python a great language to learn basic programming concepts, which can be used to learn other programming languages.



A brief intro to Python



Python is a high-level, general-purpose programming language.

Its design philosophy emphasizes code readability.

This makes Python a great language to learn basic programming concepts, which can be used to learn other programming languages.

Just like any other language, there are rules on how to write code.
We call these rules the ***syntax***.

For example, consider the role of the comma in these two sentences:

Let's eat grandma vs Let's eat, grandma

Computers need *exact* instructions, so poor syntax (e.g., typos)
will lead to code errors,

Let's look at a Python code

```
from dataclasses import dataclass
from pathlib import Path

# for compatibility across different versions of Python
try:
    # compatible with Python >= 3.11
    import tomllib
except ImportError:
    import tomli as tomllib

# import the list of possible patterns
PATTERNS_FILE = Path(__file__).parent / "patterns.toml"
```

```
# The decorator @dataclass automatically generates special methods
# like __init__ and __repr__ to user-defined classes, which saves us effort!
# see documentation: https://docs.python.org/3/library/dataclasses.html
```

```
@dataclass
```

```
class Pattern:
```

```
    """ Pattern class in Conway's Game of Life
```

```
    A square array of 'dead' and 'alive' cells is tracked at each iteration.
    In this implementation, the array is initiated with 'dead' cells and
    we keep track of coordinate index of the 'alive' cells only as a list of tuples
```

```
    """
```

```
    name: str
```

```
    alive_cells: set[tuple[int, int]]
```

To make code easier to read for humans, we often put in descriptive **comments**

Two types of comments: line comments (marked with #) and block comments (marked with """)

Let's look at a Python code

```
from dataclasses import dataclass
from pathlib import Path
```

```
# for compatibility across different versions of Python
try:
```

```
    # compatible with Python >= 3.11
    import tomllib
```

```
except ImportError:
    import tomli as tomllib
```

```
# import the list of possible patterns
```

```
PATTERNS_FILE = Path(__file__).parent / "patterns.toml"
```

```
# The decorator @dataclass automatically generates special methods
```

```
# like __init__ and __repr__ to user-defined classes, which saves us effort!
```

```
# see documentation: https://docs.python.org/3/library/dataclasses.html
```

```
@dataclass
```

```
class Pattern:
```

```
    """ Pattern class in Conway's Game of Life
```

```
    A square array of 'dead' and 'alive' cells is tracked at each iteration.
```

```
    In this implementation, the array is initiated with 'dead' cells and
```

```
    we keep track of coordinate index of the 'alive' cells only as a list of tuples
```

```
    """
```

```
    name: str
```

```
    alive_cells: set[tuple[int, int]]
```

Python is modular.
We can import
different *packages*

Let's try a little coding and simulating!

Thanks for participating!

Survey



<https://tinyurl.com/automata-survey>